

SPORTS ANALYTICS

MECE Analysis Framework

A structured framework for organizing the Olympic sports analytics project into five mutually exclusive and collectively exhaustive analytical areas.

PROJECT OBJECTIVE

To analyze historical Olympic data and identify meaningful patterns and trends across Olympic Games, sports and events, athlete participation, country and regional representation, and medal performance using EDA, SQL, and Power BI.

MECE at a Glance

Analytical Area	Primary Focus
01 Olympic Games Overview	Games, seasons, historical development and host cities
02 Sports & Events	Sports, events, gender categories and event evolution
03 Athlete & Participation	Athletes, demographics, age and participation patterns
04 Country & Regional Analysis	NOCs, countries, regions and geographical representation
05 Medal & Performance	Medal outcomes, leading performers, sports and regional performance

01 Olympic Games Overview

Objective: Analyze the historical development, structure, frequency and hosting patterns of the Olympic Games.

1. Total Olympic Games
2. Games by Season (Summer vs. Winter)
3. Games by Decade
4. Total Host Cities
5. Games Hosted by City

Key analysis questions

- How have Olympic Games changed across seasons and decades?
- Which cities have hosted the Games most frequently?
- What historical patterns are visible in Olympic editions?

02 Sports & Events

Objective: Understand the composition, diversity and evolution of Olympic sports and events.

1. Total Sports
2. Total Events
3. Sports by Season
4. Events by Sport

5. Events by Gender

Key analysis questions

- How do sports and events differ between Summer and Winter Olympics?
- Which sports contain the largest number of events?
- How have events and sports participation evolved over time?

03 Athlete & Participation

Objective: Analyze athlete participation, demographic characteristics and participation patterns over time.

1. Total Participants

2. Participants by Gender

3. Average Age

4. Participation by Sport

5. Participation Trends Over Time

Key analysis questions

- How does participation vary by gender, age and sport?
- How has athlete participation changed over time?
- Which participant characteristics and sports show notable patterns?

04 Country & Regional Analysis

Objective: Understand geographical representation and participation across countries, NOCs and regions.

1. Total NOCs/Regions

2. NOCs/Regions by Olympic Games

3. Participants by Country/Region

4. Top Countries/Regions by Participation

5. Regional Participation Trends

Key analysis questions

- Which countries and regions have the highest participation?
- How has the number of participating NOCs/regions changed over time?
- What geographical patterns are visible in Olympic participation?

05 Medal & Performance

Objective: Evaluate Olympic performance through medal distribution across Games, sports, countries and regions.

1. Total Medals

2. Gold / Silver / Bronze Medals

3. Medals by Olympic Games

4. Medals by Sport

5. Medals by Country/Region

Key analysis questions

- How are medals distributed across Games, sports and medal types?

- Which countries and regions have the strongest medal performance?
- Which sports account for the largest medal totals?

Dashboard KPI Direction

The MECE framework defines the broader analytical scope. The overview dashboard should remain visually clean and use only five headline KPIs, while supporting charts provide the detailed breakdowns.

#	Headline KPI	MECE Area
1	Total Olympic Games	Olympic Games Overview
2	Total Sports	Sports & Events
3	Total Events	Sports & Events
4	Total Participants	Athlete & Participation
5	Total Medals	Medal & Performance

Conclusion

The MECE framework provides a structured and comprehensive approach to analyzing Olympic data across five key dimensions: **Olympic Games, Sports & Events, Athletes & Participation, Country & Regional Analysis, and Medal & Performance**. By combining EDA, SQL, and Power BI, the analysis will help identify important patterns, trends, participation characteristics, geographical representation, and performance outcomes throughout Olympic history.

This framework ensures that the analysis remains **focused, non-overlapping, and data-driven**, while providing a clear foundation for the final dashboard, report, and presentation.